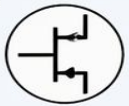
FIELD EFFECT TRANSISTORS (FETs)

JFET & MOSFET: Characteristics, Parameters, Biasing

1. WHAT IS A FET?

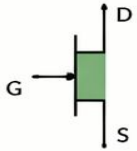
A Field Effect Transistor (FET) is a voltage-controlled device. It uses an electric field to control current flow between two terminals.



- **High input impedance**
- **Low power consumption**

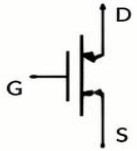
2. TYPES OF FETs

JFET



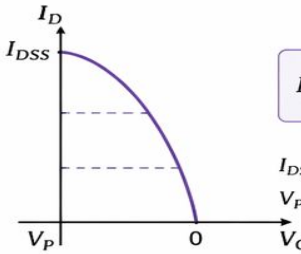
- Junction Gate Field Effect Transistor

MOSFET



- Metal Oxide Semiconductor FET

3. JFET CHARACTERISTICS

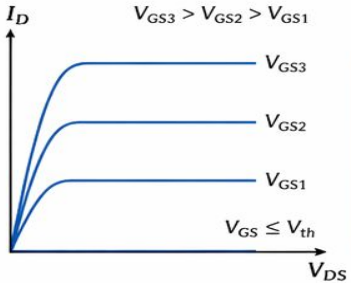


$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

I_{DSS} – Drain current at $V_{GS} = 0$

V_P – Pinch-off (cut-off) voltage

4. MOSFET CHARACTERISTICS



Regions:

- Cut-off: $V_{GS} < V_{th}$
- Triode (Linear): $V_{DS} < V_{GS} - V_{th}$
- Saturation: $V_{DS} \ge V_{GS} - V_{th}$

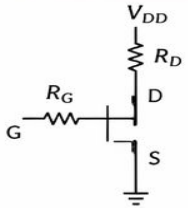
V_{th} – Threshold voltage

5. IMPORTANT PARAMETERS

I_{DSS}	Drain current at $V_{GS} = 0$ (JFET)
V_P	Pinch-off voltage (JFET)
V_{th}	Threshold voltage (MOSFET)
g_m	Transconductance
r_{ds}	Output resistance
R_{in}	Input resistance (very high)

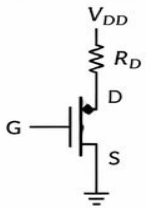
6. BIASING CONFIGURATIONS

JFET Biasing



Gate is usually reverse biased (by R_G).

MOSFET Biasing



Gate is biased by a voltage divider.

7. QUESTIONS & EXERCISES

- Draw and explain the transfer characteristic of a JFET.
- Compare JFET and MOSFET.
- Determine I_D for a given V_{GS} in a JFET.
- Design a biasing circuit for a MOSFET and calculate operating point.

Introduction to Field Effect Transistors

Field Effect Transistors (FETs) are semiconductor devices used to control the flow of electric current in electronic circuits. They are **voltage-controlled devices**, meaning the current between the **Drain (D)** and **Source (S)** terminals is controlled by the voltage applied to the **Gate (G)** terminal.

FETs are widely used in electronics because they have **high input impedance**, **low power consumption**, and good switching performance. They are commonly used in amplifiers, switching circuits, and integrated circuits.

Types of Field Effect Transistors

There are two main types of Field Effect Transistors:

JFET (Junction Field Effect Transistor)
JFET controls current using a **reverse-biased PN junction**. The gate voltage changes the width of the channel, which controls the current flowing between the drain and source.

MOSFET (Metal Oxide Semiconductor Field Effect Transistor)
MOSFET has an **insulated gate separated by a thin oxide layer**. Because of this structure, MOSFETs have extremely **high input resistance** and are widely used in modern digital circuits and microprocessors.

JFET Characteristics

In a JFET, the drain current depends on the **gate-source voltage (VGS)**. When the gate voltage becomes more negative, the channel becomes narrower and the current decreases.

The relationship between drain current and gate voltage is given by:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

Where:

- **IDSS** – Maximum drain current when VGS=0
- **VP** – Pinch-off voltage

When the gate voltage reaches the pinch-off value, the current becomes almost zero.

MOSFET Characteristics

MOSFET operation is mainly controlled by the **threshold voltage (Vth)**. A conducting channel is created only when the gate voltage exceeds this value.

MOSFET operates in three regions:

Cut-off region

$V_{GS} < V_{th}$ → Transistor is OFF

Triode (Linear) region

$V_{DS} < V_{GS} - V_{th}$ → Acts like a variable resistor

Saturation region

$V_{DS} \geq V_{GS} - V_{th}$ → Used for amplification

Important Parameters of FETs

Several parameters describe the performance of FET devices:

- **IDSS** – Drain current at zero gate voltage (JFET)
- **VP** – Pinch-off voltage
- **Vth** – Threshold voltage (MOSFET)
- **gm (Transconductance)** – Shows how effectively the gate voltage controls the drain current
- **rds** – Output resistance of the transistor

These parameters are important when designing and analyzing electronic circuits.

Biasing of FETs

Biasing is the process of setting a stable **operating point (Q-point)** for the transistor so that it works properly in a circuit.

In **JFET biasing**, the gate is usually **reverse biased** using a gate resistor.

In **MOSFET biasing**, the gate voltage is often set using a **voltage divider network**. Proper biasing ensures stable operation and prevents distortion in amplifier circuits.